Mobile App Development

Programming languages are engineered tools used to translate human intent into machine-readable instructions'

Two types of programming language:

Low-Level Languages: These are machine-friendly languages that offer direct control over hardware and memory but are incredibly difficult for humans to read.

High-Level Languages: Human-readable languages that use words resembling English and standard math symbols. They are highly portable across different operating systems but require translation via compilers or interpreters to run. Examples include **Python, Java, and C++ Javascript**

Execution Type	How it Works	Pros & Cons	Examples
Compiled	A compiler translates the entire source code into a standalone machine code file *before* running it.	➕ Extremely fast execution ➖ Must re-compile every time code changes	C, C++, Rust, Go
Interpreted	An interpreter reads and executes the code line by line, on the fly, at runtime.	➕ Flexible, easy to debug, platform-independent ➖ Slower execution speed	Python, JavaScript, Ruby, PHP

Coding

Coding is the process of writing step-by-step instructions that tell a computer how to perform specific tasks.

Communication: Humans cannot talk directly to computers in everyday languages, and computers only understand basic electrical signals or binary (0s and 1s).

Execution: Once the computer translates the instructions, it runs the code to complete an action

// Selecting an HTML element

const actionButton = document.querySelector('button'); // Adding interactive behavior actionButton.addEventListener('click', () => { actionButton.textContent = 'You clicked me!'; });

JavaScript is a lightweight programming language primarily used to add interactivity and dynamic behavior to websites. Along with HTML (which structures content) and CSS (which styles it), JavaScript forms one of the three core building blocks of the World Wide Web. Without it, web pages would be completely static—meaning they could only display text and images without responding to your actions

Words = keywords

Sentence = statement